USN	0	2	J	S	T	2	4	V	C	S	0	3	0	Course Code:	24CS340

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester B.E Degree Examination

Department of Computer Science and Engineering

COMPUTER ORGANIZATION

Duration: 3 Hours | Max. Marks: 100

Instructions:

- Answer all the questions.
- Part- B has internal choice.

PART – A

Q.NO	1	CO1	1	CD	L3	PI	2.2.3	MARKS	5+5

a. Explain the various registers available in the processor of a computer and describe how the processor is connected to memory.

b. What are condition code flags? Explain the four commonly used flags.

Q.NO	2	CO2	2	CD	L2	PI	1.4.1	MARKS	10

Describe in detail the interrupt mechanism? How can interrupts be enabled or disabled? Also, list the sequence of events involved in handling an interrupt request from a single device.

Q.NO	3	CO3	3	CD	L2	PI	1.4.1	MARKS	10

Discuss the implementation and working of static RAM and dynamic RAM cells, with suitable diagrams.

Q.NO	4	CO4	4	CD	L3	PI	2.2.3	MARKS	5+5

a. Explain the generate G_i and propagate P_i functions in a carry look-ahead adder for stage i, along with the bit-stage cell diagram.

b. Show the multiplication of 13 (multiplicand) and -11 (multiplier) using multiplier bit pair recoding technique.

Q.NO	5	CO5	5	CD	L2	PI	1.4.1	MARKS	5+5

a. List the actions needed to execute the instruction Load R5, X(R7).

b. Illustrate the conceptual hardware organization involving the register file and the ALU used for performing computations.

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PART – B

Q.NO	6	CO1	1	CD	L3	PI	2.2.3	MARKS	5+5

a. Register R5 is used in a program to point to the top of a stack containing 32-bit numbers. Write a sequence of instructions using the Index, Autoincrement, and Autodecrement addressing modes to perform each of the following tasks:
   i) Pop the top two items off the stack, add them, then push the result onto the stack.
   ii) Copy the fifth item from the top into register R3.
   iii) Remove the top ten items from the stack.
   For each case, assume that the stack contains ten or more elements.

b. Explain the process of assembling and executing an assembly language program.

OR

Q.NO	7	CO1	1	CD	L3	PI	2.2.3	MARKS	5+5

a. Represent the decimal values 43, -10 and -27 in the following 7-bit binary formats.
   i) sign-and-magnitude ii) 1's complement iii) 2's complement

b. What are stack frames? Discuss their role in the implementation of subroutines.

Q.NO	8	CO2	2	CD	L2	PI	1.4.1	MARKS	10

How does program-controlled I/O operate and why does it require continuous processor monitoring of I/O devices? Explain with a keyboard-to-display data transfer example.

OR

Q.NO	9	CO2	2	CD	L2	PI	1.4.1	MARKS	4+6

a. Describe the processor control registers that support interrupt processing.

b. Explain the timing of an input transfer on an asynchronous bus with a timing diagram.

Q.NO	10	CO3	3	CD	L2	PI	1.4.1	MARKS	10

Explain the organization of cache memory using fully associative mapping with a neat illustration. Also discuss how it improves the memory access time and how it is different from set associative mapping?

OR

Q.NO	11	CO3	3	CD	L3	PI	2.2.3	MARKS	5+5

a. Consider a memory consisting of 64K words of 8 bits each. Discuss how this memory can be organized using 16K X 1 static memory chips.

b. With a block diagram, explain the virtual memory organization. Also list the benefits of a virtual memory system.

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Q.NO	12	CO4	4	CD	L3	PI	2.2.3	MARKS	5+5
a. Divide 1110 by 0011 using restoring division. b. Explain the IEEE standards for floating point number representation.									
OR									
Q.NO	13	CO4	4	CD	L2	PI	1.4.1	MARKS	10
With a neat diagram, explain the floating point addition/subtraction unit.									

Q.NO	14	CO5	5	CD	L2	PI	1.4.1	MARKS	10
Discuss the 5-stage processor datapath with the help of a neat diagram.									
OR									
Q.NO	15	CO5	5	CD	L2	PI	1.4.1	MARKS	10
With a neat diagram, explain the hardwired control unit.									

Course Outcome (CO):	
CO–1	Apply the basic concepts of computer, instruction set architecture, function and characteristics of a computer system and design the assembly language for the sequence of machine instruction.
CO–2	Explain the different ways of communicating with the I/O devices.
CO–3	Design and analyse the performance of memory systems.
CO–4	Apply the Arithmetic and Logical operations with integer and floating–point operands.
CO–5	Illustrate the basic processing unit and organization of simple processor.

PI	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem.
2.2.3	Identify existing solution/methods to solve the problem, including forming justified approximations and assumptions.

Cognitive Domain (CD):	
L1	Remember
L2	Understand
L3	Apply

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